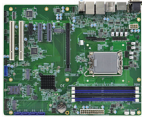


I/O interfaces for connectivity. The motherboard offers various wireless connectivity options, including 5G/4G/



LTE connectivity with a 5G-compatible M.2 B3052 socket and a discrete Wi-Fi 6E.

MBB-1000 ATX uses long-term stable capacitors and durable components combined with stringent testing procedures to meet mission-critical computing needs. The platform is suitable for an industrial platform with long-term reliability for graphics-intensive applications, such as AIoT, digital signage, medical imaging, video surveillance, and interactive kiosks.

IBASE
www.ibase.com.tw

AVR-IoT cellular development board

The AVR-IoT cellular development board from Microchip is designed to ease the process of embedded design. The mini development board is based on an 8-bit



AVR128DB48 MCU. The module is capable of providing a robust platform to start building sensor and actuator nodes on 5G narrowband IoT networks. The small form factor and security protection offered by Microchip's ATECC608 crypto authentication device make it a preferable solution for developers who want to connect IoT devices to an available 5G network.

Microchip
<https://www.microchip.com>

5G-ready IoT module

SARA-R500E is u-blox's first module with an onboard embedded SIM (eSIM). It is based on u-blox's UBX-R5 cellular chipset and is designed for space-constrained IoT applications. SARA-R500E is 5G-ready and employs the SARA-R5 AT command interface.



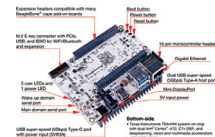
Moreover, it is pin-to-pin compatible with other modules in the SARA family.

The elimination of a removable SIM card allows the product designers to design a compact, robust, and a tightly sealed device that can meet the demanding IP67 and IP68 criteria. The module is ideal for rugged IoT applications employed in harsh environments, such as surveillance cameras, asset trackers, and smart meters.

u-blox
www.u-blox.com

64-bit single-board computer

BeagleBone AI-64 SBC is a 64-bit open hardware single-board computer for embedded applications. It is powered by a Texas Instruments TDA4VM SoC



with a dual Arm Cortex-A72 core. The board features a programmable C7x DSP core, deep learning, vision and multimedia accelerators. The board offers multiple I/O interfaces, including USB-3 Type C, Mini DisplayPort interface, and 4-Lane CSI connector for camera options. The computer is suitable for use in a variety of applications, which include autonomous robots and drones, home security, media servers, machine learning, machine vision, and computer vision.

BeagleBone
<https://beagleboard.org>

T&M INSTRUMENTS

Portable mixed signal oscilloscope

Tektronix has released 2 Series mixed signal oscilloscope (MSO). The multifunctional instrument features



an optional arbitrary function generator, pattern generator, voltmeter, and frequency counter. The portable oscilloscope weighs about 1.8kg and is 3.8cm inches thin. The 2 Series MSO includes TekDrive, a test and measurement data workspace in the cloud where engineers can upload, store, organise, and share any file

from a connected device. It comes with a power backup of up to eight hours.

Tektronix
<https://www.tek.com/en>

Metal composition analyser

Spectro Analytical Instruments has introduced SPECTROMAXx LMX10 ARC/SPARK OES analyser that provides a faster and more accurate metal



composition data. The device offers higher reliability in material control analysis and provides precise results

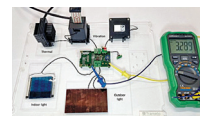
for in-process to final quality inspection, thus providing certainty to critical supply chain. The analyser comes in two models, basic and advanced, with advanced model having an additional capability to analyse oxygen in copper.

Spectro Analytical Instruments
<https://www.spectro.com>

BOARDS AND MODULES

Demo module for energy harvesting

Trameto TM2040 demonstration module can enable engineers to evaluate the power generated through micro-energy harvesting and predict the size of the battery required for an application.



It can be connected with up to four energy harvesters of the same or mixed types. Furthermore, the module is capable of powering an IoT device, such as sensors and the wireless modules, with a 1.8V DC regulated supply at up to 15mA. The platform comes with a Windows application that can display the data of the platform via a graphical user interface.

Trameto
<https://trameto.com>

Plug-and-play DC/DC converter

RMOD500-W from Recom is a compact, plug-and-play 500W DC/DC converter for off-highway E-mobility applications. The DC-DC converter can accept a wide range of input power from 32V to